

draft4 — figures-only contact sheet

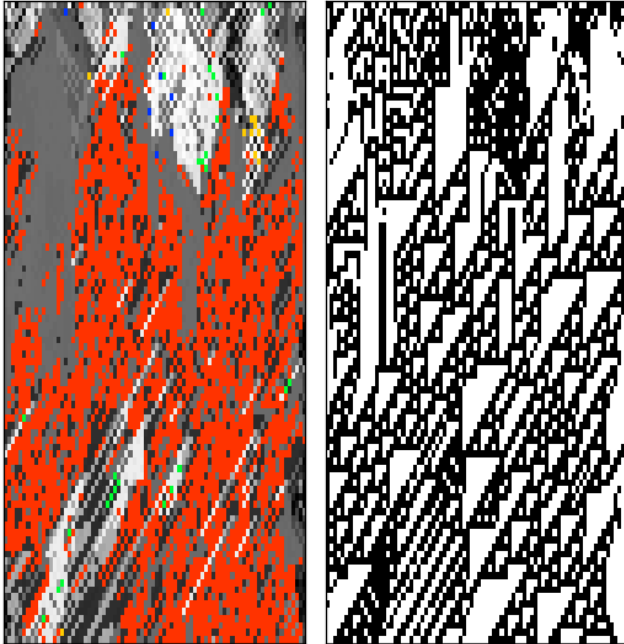
Standard Wolfram-order rebuild; every figure reproduced from `mmca-clj` alone.

July 21, 2026

`rule110_spread.png`

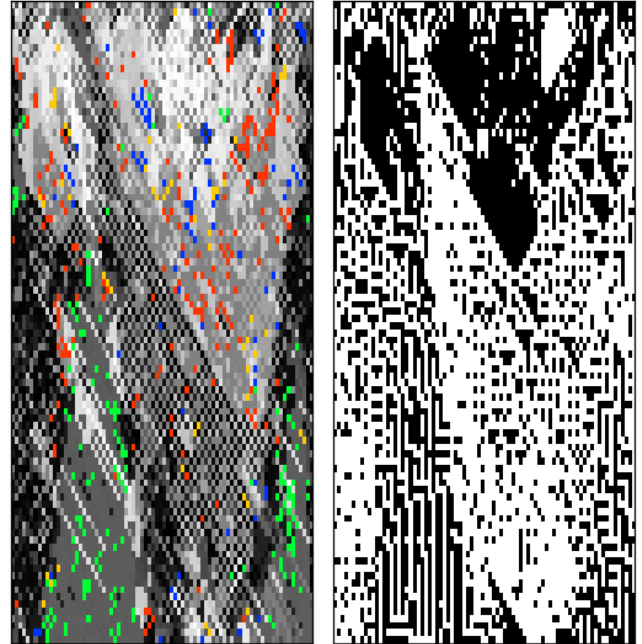
Fig 1 — Rule 110 predominates ($\sigma=16250374$, Wolfram conjugate $\approx 35\%$) vs a diverse regime (10275364)

$\sigma = 16250374$
Rule 110 predominates



Rule 110 = 39% of genotype cells (red); seed 0

$\sigma = 10275364$
a different regime — green/blue rule-domains



< 2% Rule 110; seed 0

fig8.png

Fig 2 — the “Figure 8” triptych: genotype / phenotype / population

Figure 8 (seed 4): the cast dies off; two survivors take over

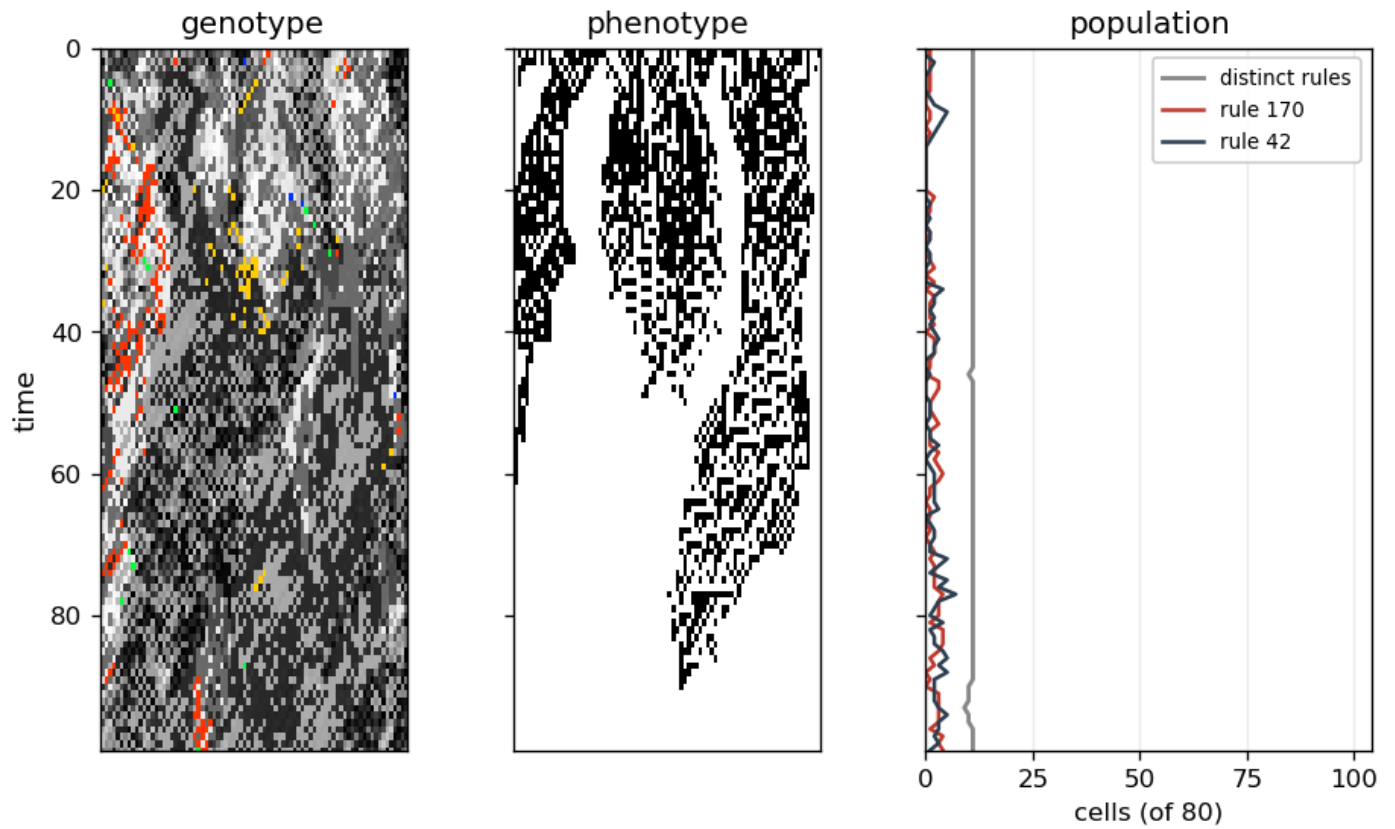
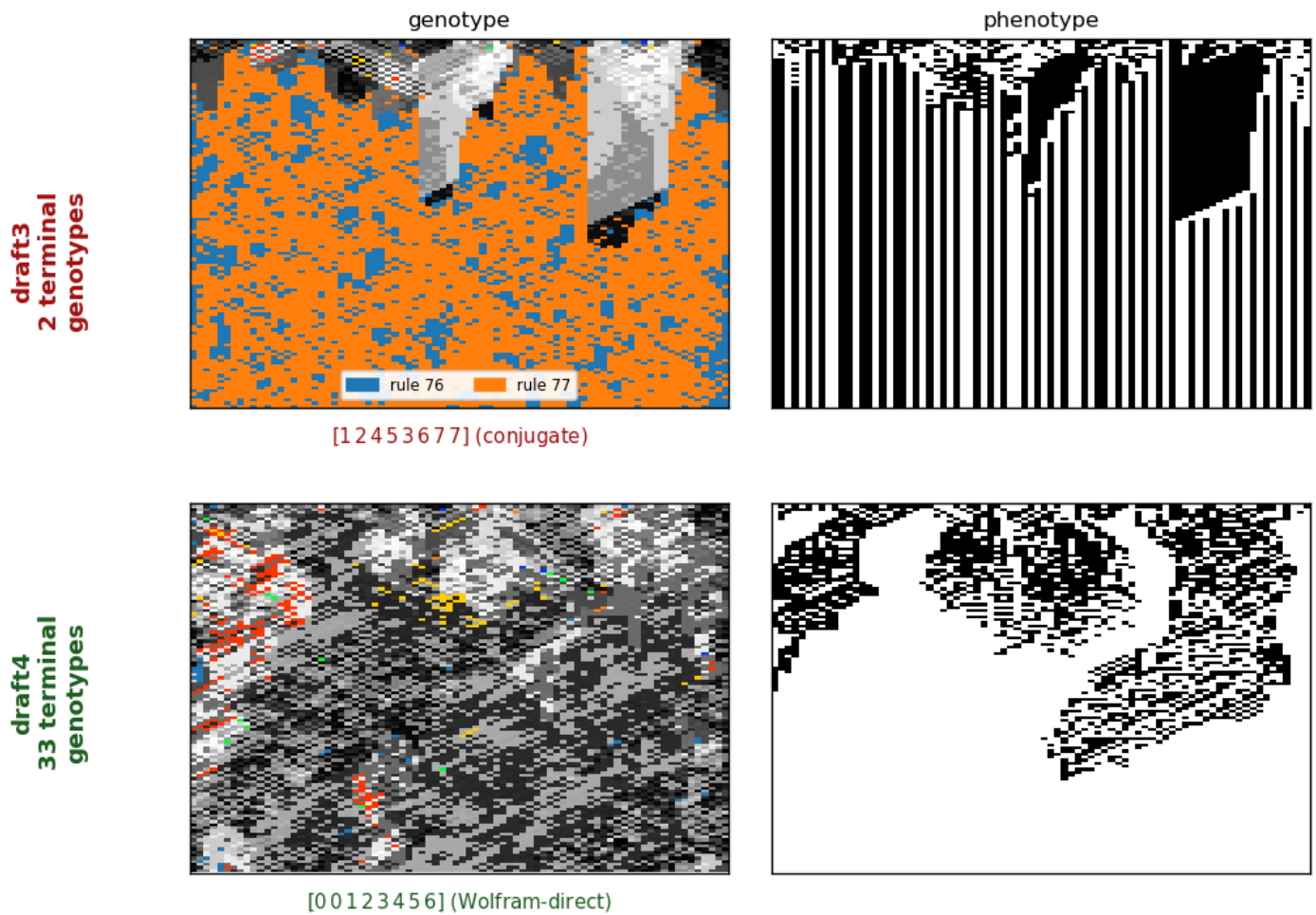


fig2pair.png

NEW — Fig 2's two terminal-diversity solutions: draft3 conjugate [1 2 4 5 3 6 7 7] dies to 2 genotypes (rules 76,77); draft4 [0 0 1 2 3 4 5 6] stays diverse (~30)

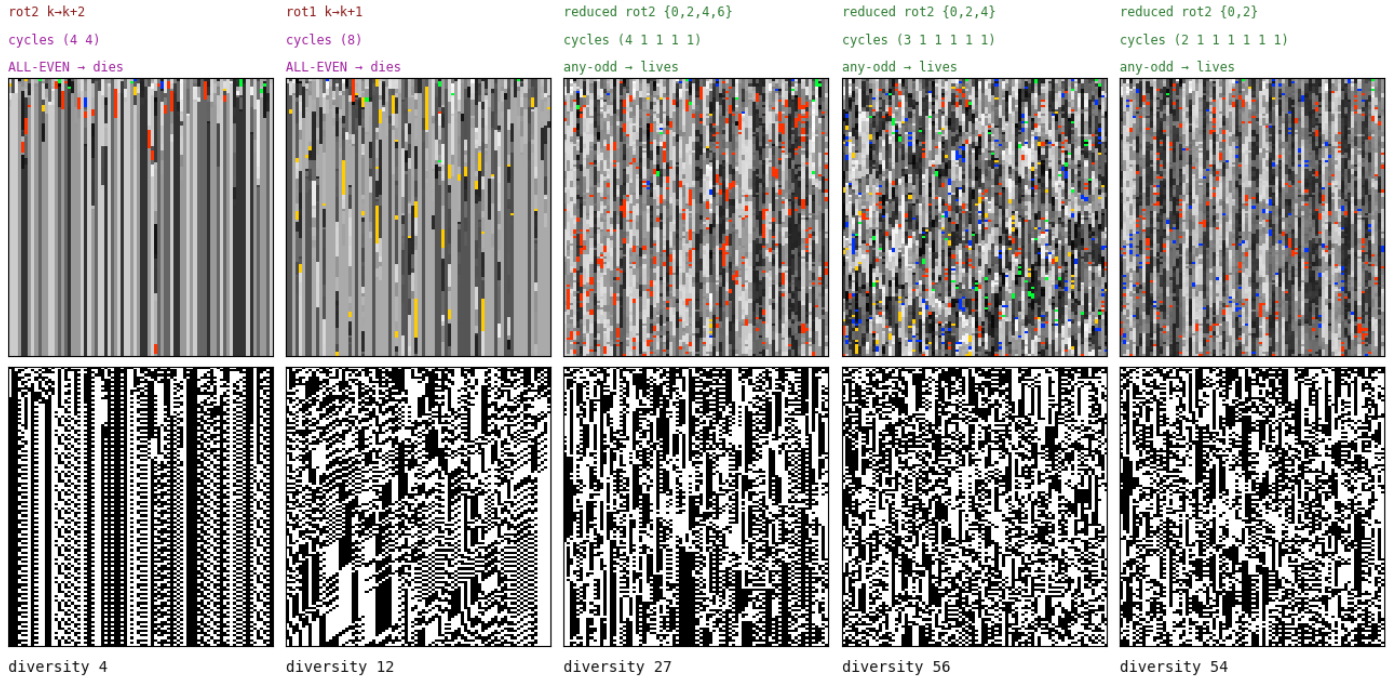
Figure 2 — two terminal-diversity solutions of the same operator under the Wolfram order:
the draft3 conjugate dies to two genotypes
the draft4 form stays diverse



parity-dichotomy.png

Fig 3 — cycle-parity dichotomy: all-even collapses, any-odd survives (run alone)

CYCLE PARITY DICHOTOMY: a Boolean fixed point exists iff every cycle is even (Lean: bool_fixed_exists_iff_hasAlternatingColouring)



survey_a.png

Fig 4a — rotation survey, negative offsets: the 8-cycles ($\pm 1, \pm 3$) sustain

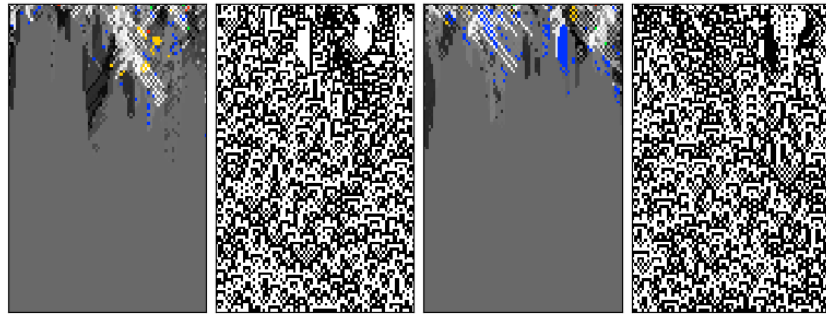
A family of operators, ranked by cycle structure (a · negative offsets)

Each row: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$, $\sigma = \text{rotate-by-offset (wrap)}$. genotype 256-colour | phenotype b/w, 100 gens, two seeds.

offset -4

gcd 4 · four 2-cycles

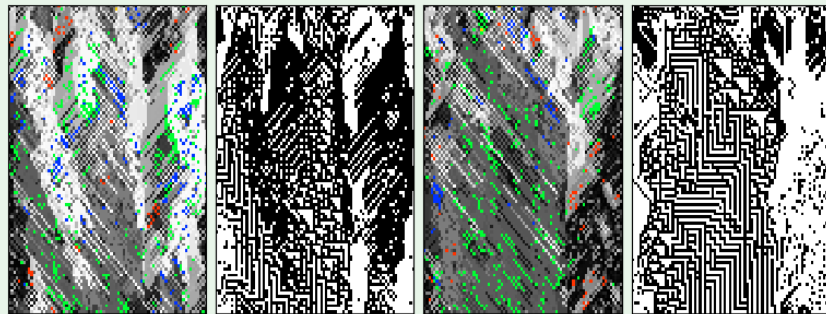
collapses ~t45



offset -3

gcd 1 · one 8-cycle

no collapse (~t100)



offset -2

gcd 2 · two 4-cycles

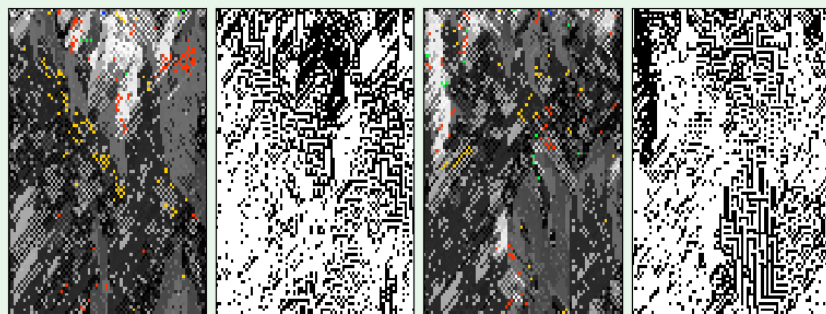
collapses ~t28



offset -1

gcd 1 · one 8-cycle

no collapse (~t100)



survey_b.png

Fig 4b — rotation survey, positive offsets

A family of operators, ranked by cycle structure (b · positive offsets)

Each row: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$, $\sigma = \text{rotate-by-offset (wrap)}$. genotype 256-colour | phenotype b/w, 100 gens, two seeds.

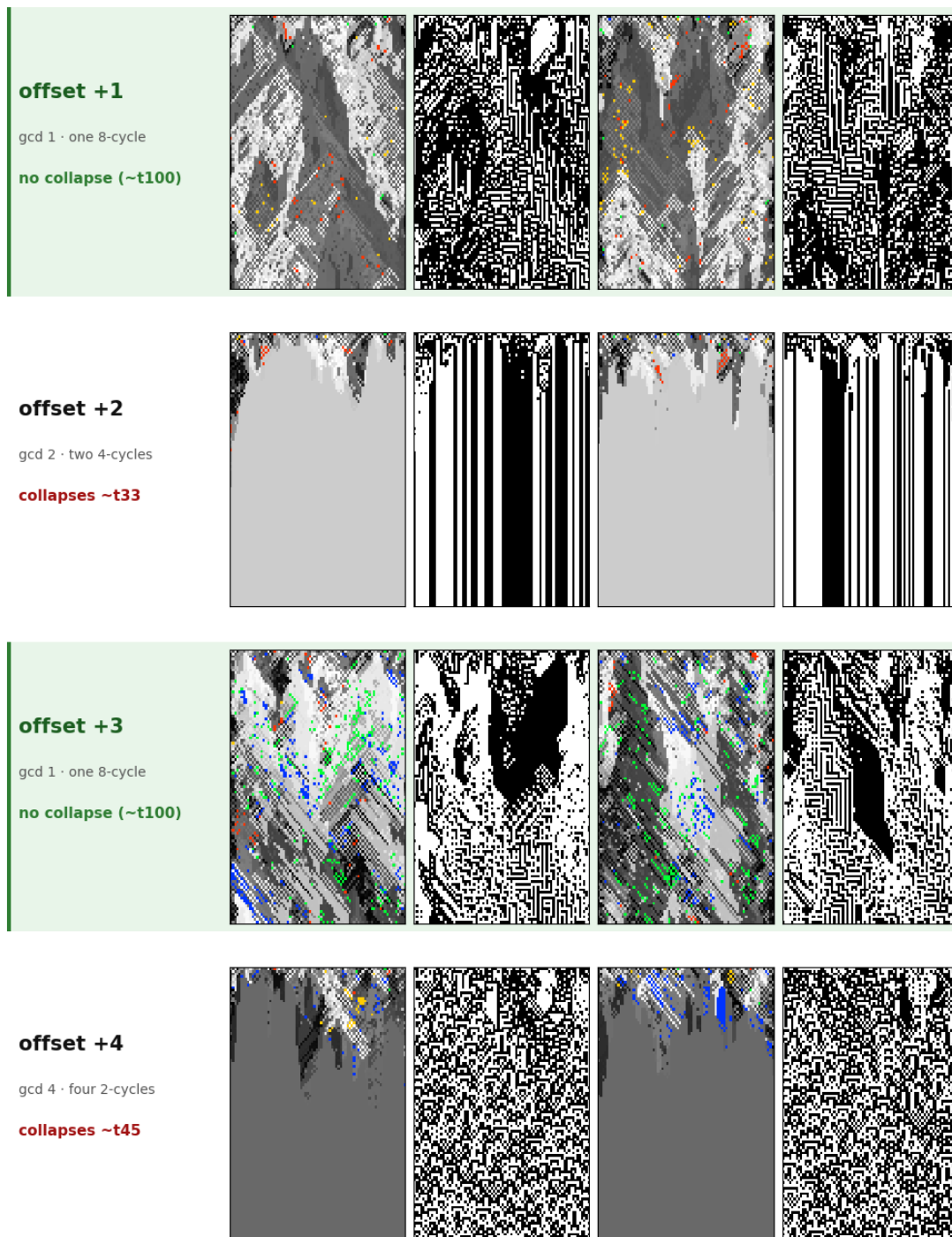
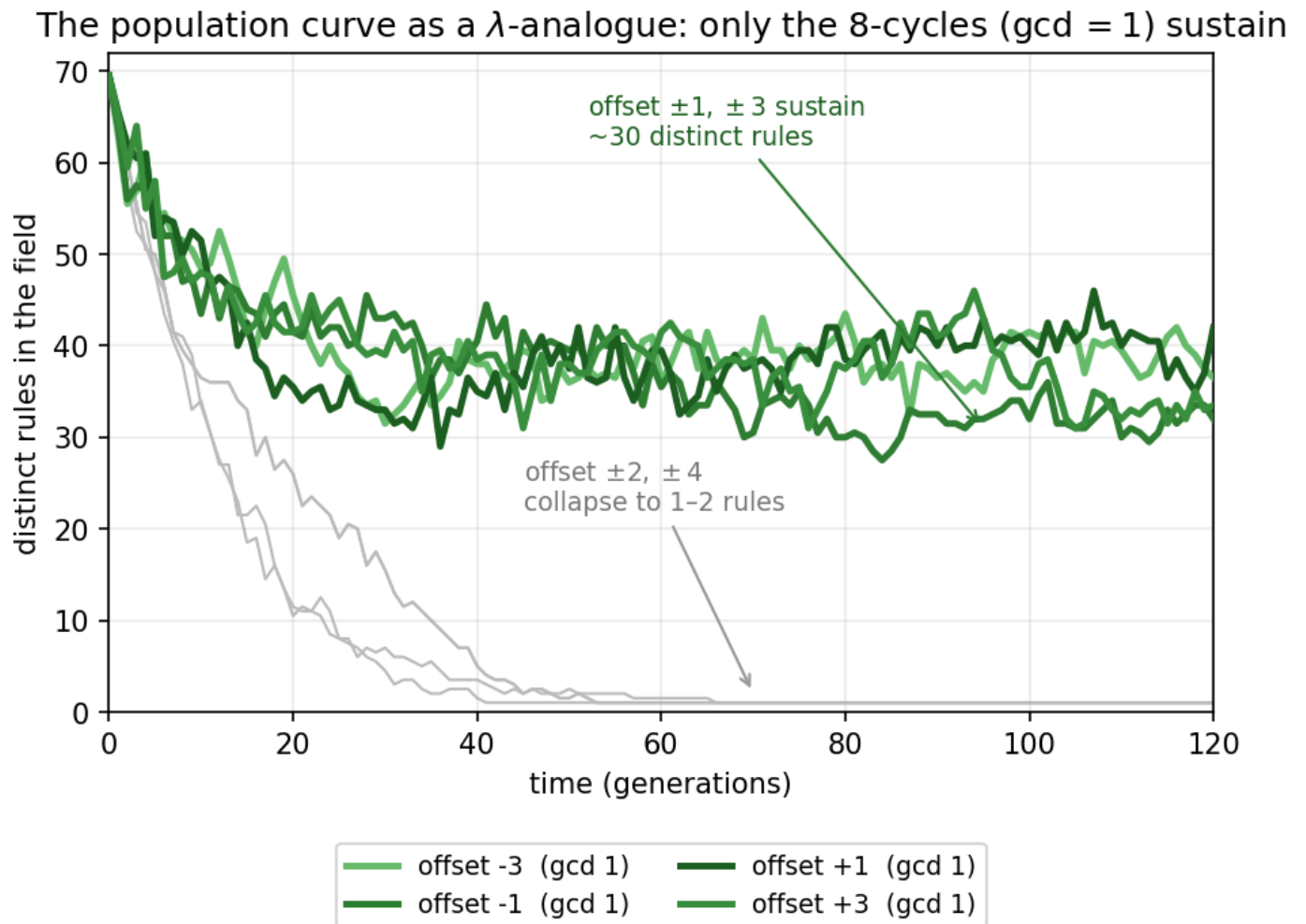


Fig 5 — population curve as λ -analogue: only $\text{gcd} = 1$ (8-cycles) sustain

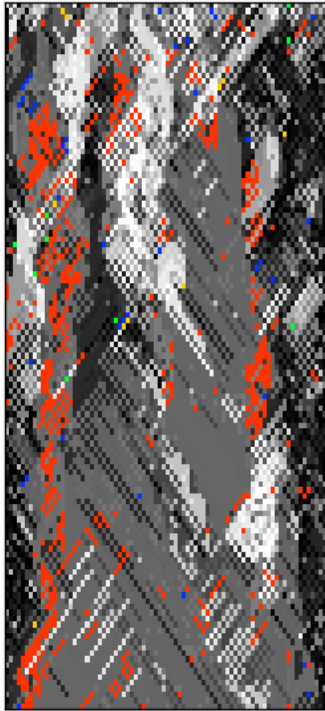
two4cycle.png

NEW — two 4-cycles, opposite fate: cycle structure does not determine aliveness (left = draft3 offset+2's Wolfram conjugate, sustains; right = Wolfram offset+2, collapses)

Same cycle type (two 4-cycles), opposite fate: cycle structure does not determine aliveness

[67021435] — **two 4-cycles**

sustains (live / live)
genotype



phenotype

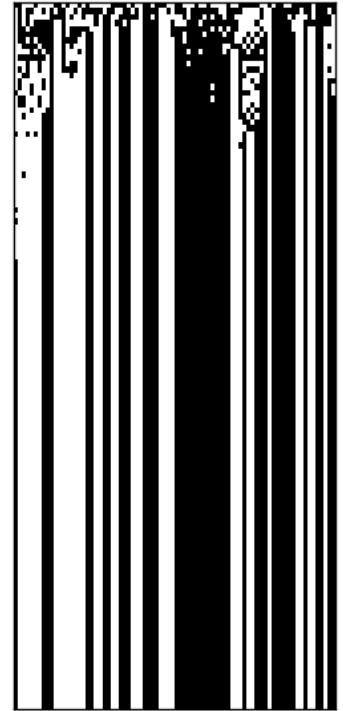


[23456701] = **offset +2** — **two 4-cycles**

collapses (dead / dead)
genotype



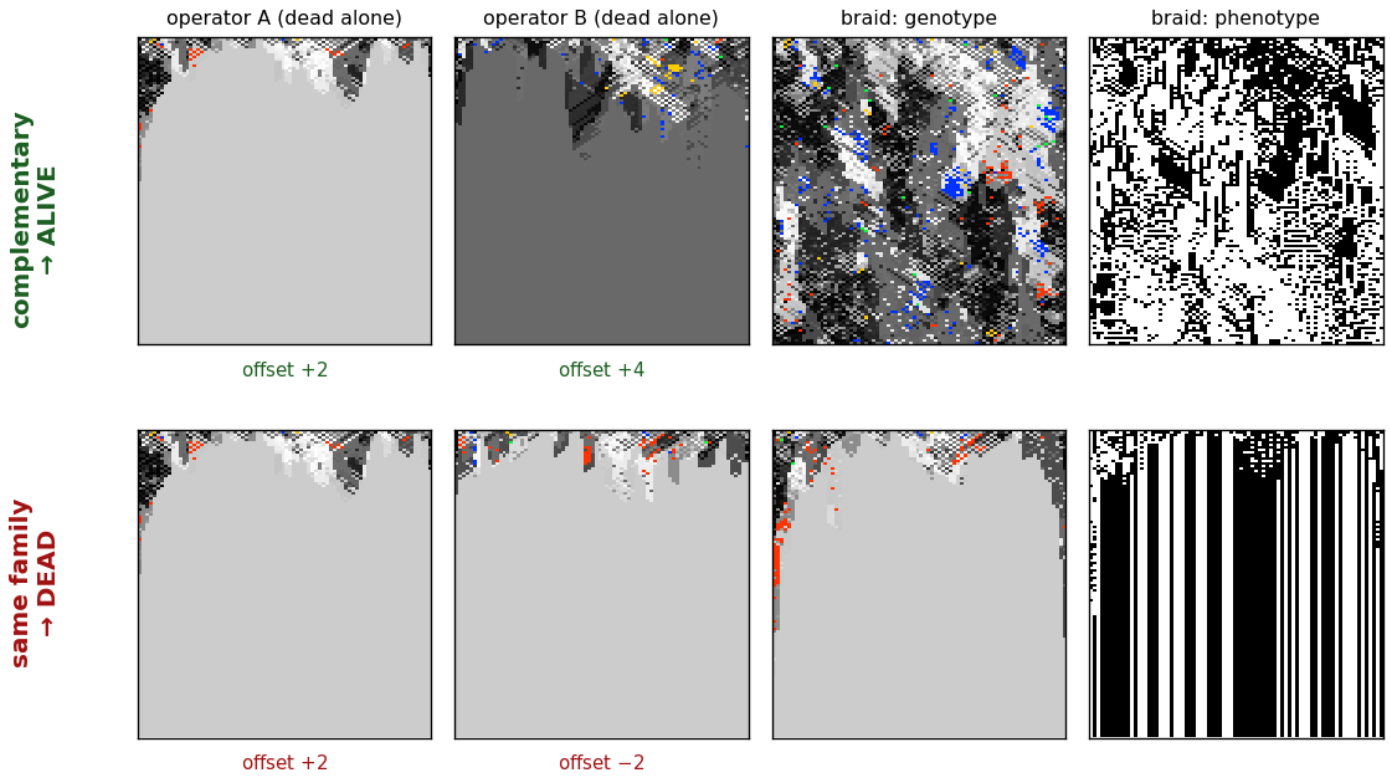
phenotype



braid.png

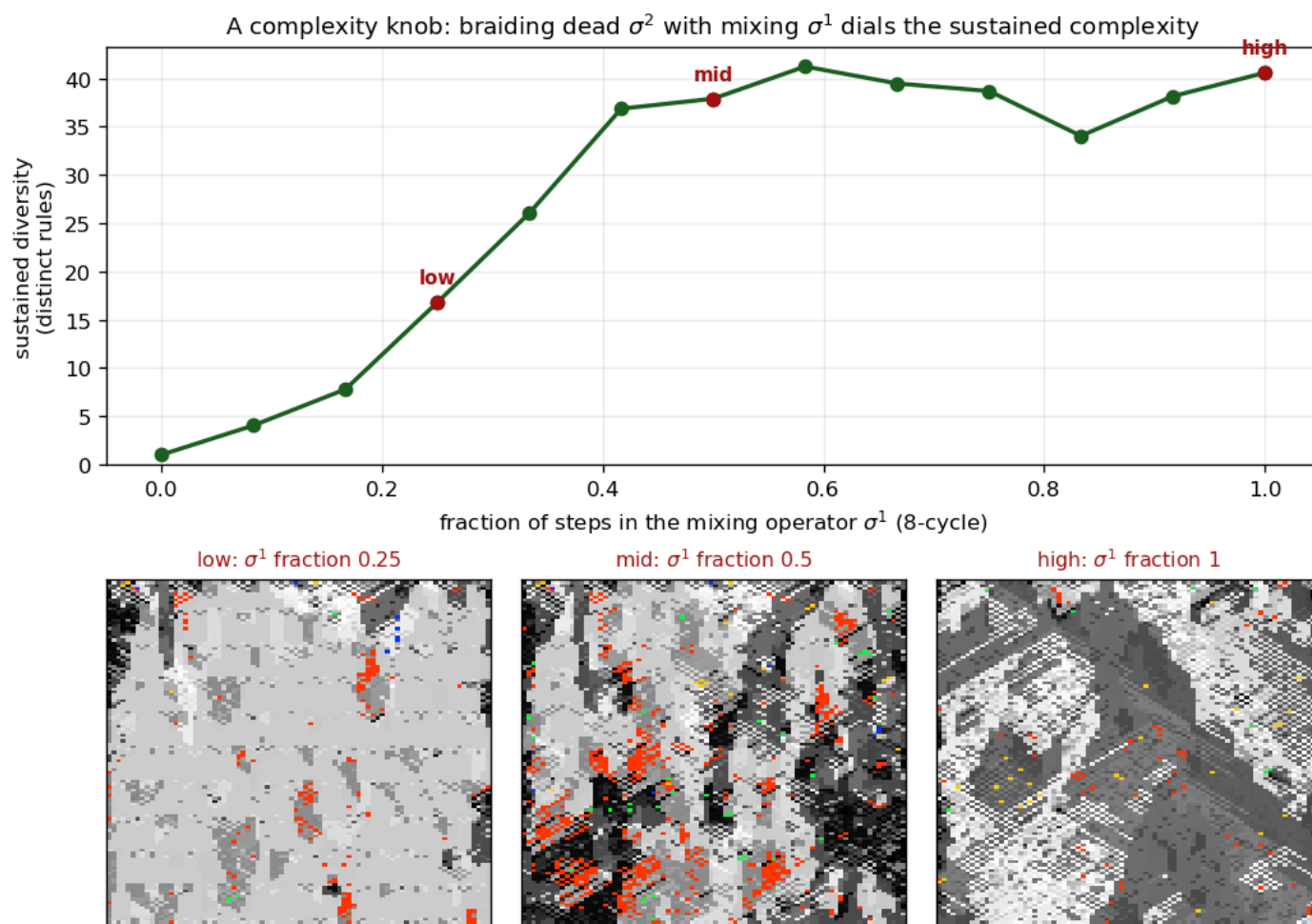
NEW — braiding two operators that each collapse alone: $\text{offset}+2 \times \text{offset}+4$ (different block systems) revives; $\text{offset}+2 \times \text{offset}-2$ (same partition) stays dead

Braiding two operators that each collapse alone: complementary block systems revive, the same partition stays dead

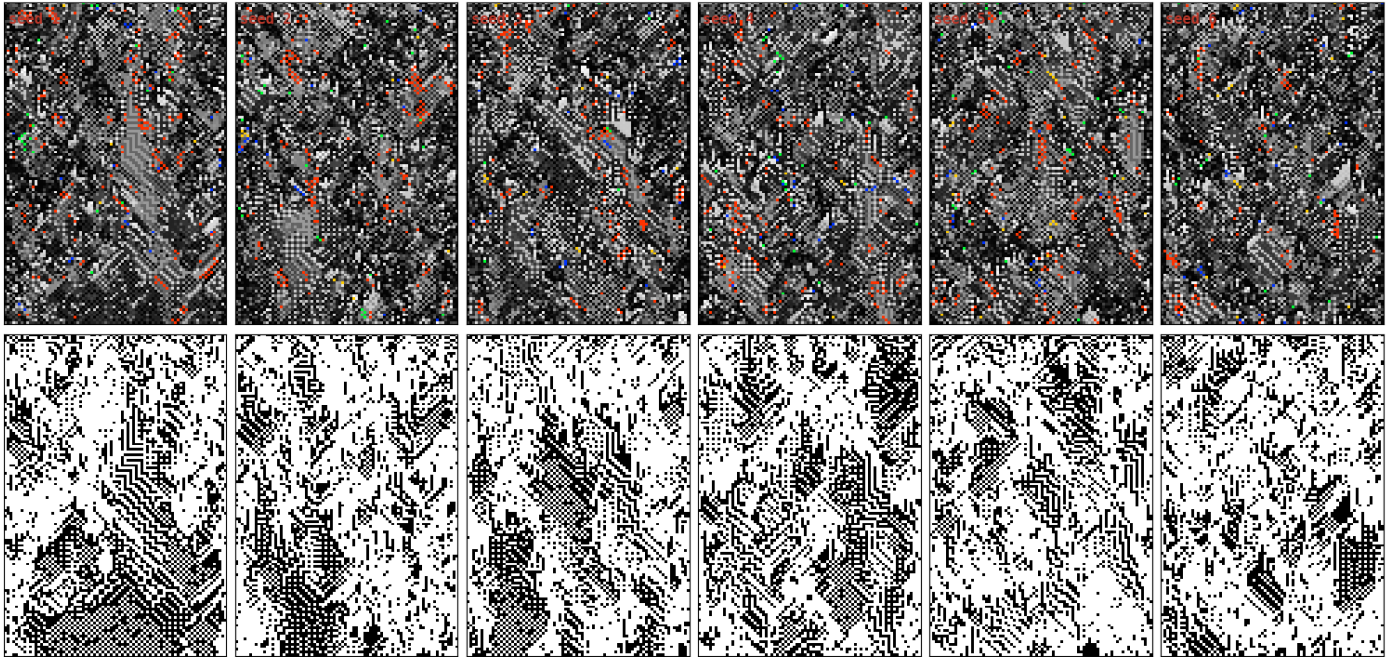


knob.png

NEW — complexity knob: braiding dead σ^2 with mixing σ^1 (powers of an 8-cycle) dials sustained diversity from dead to full;
panels show the field thickening low→mid→high



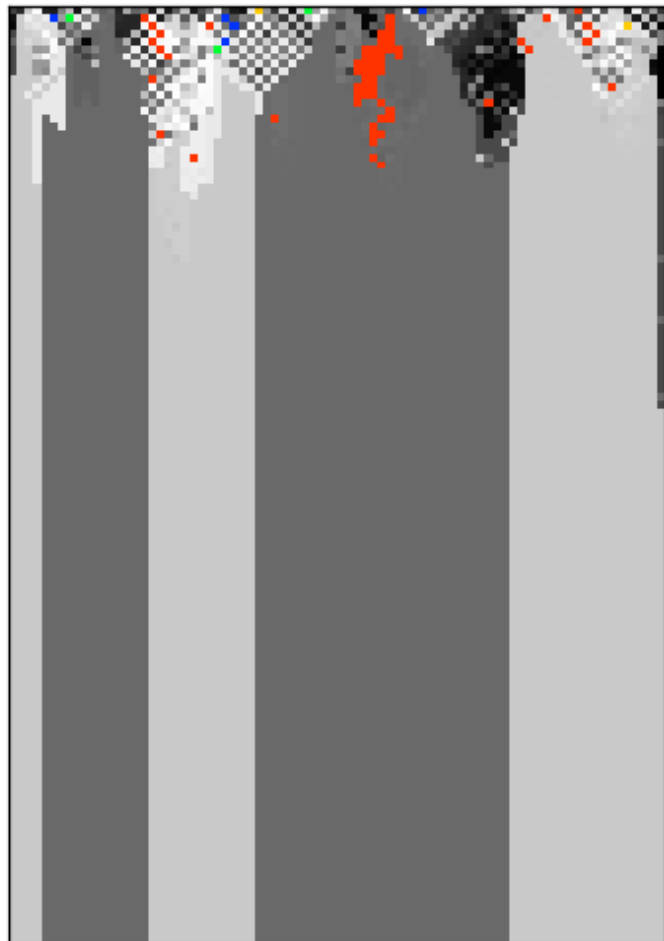
river.png
Fig 6 — the river



critical_shell.png

Fig 7 — a dead genotype (collapses to Rule 105) carrying a live phenotype

genotype — collapses to Rule 105 (dead)



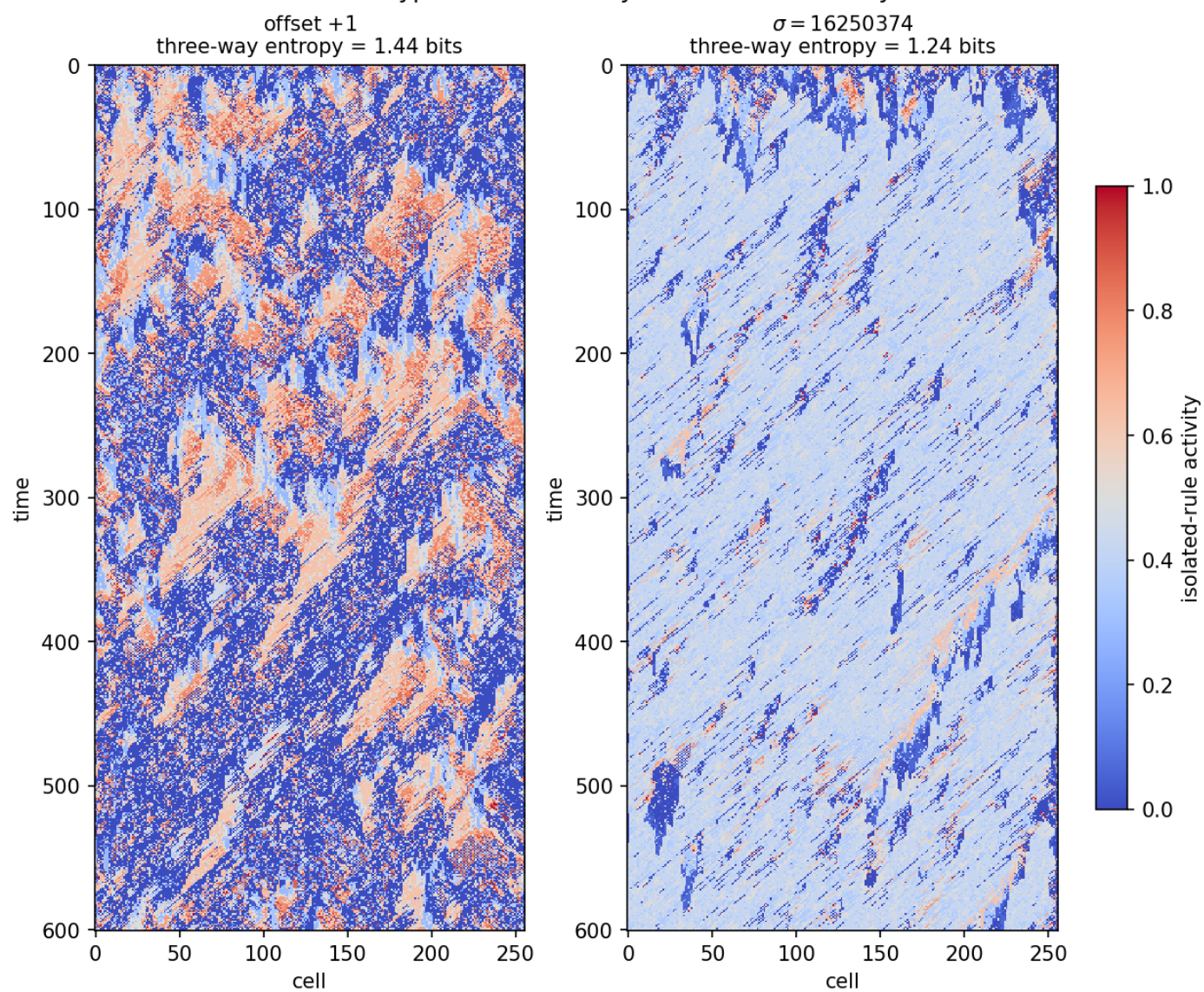
phenotype — stays complex (alive)



eoc_tint.png

EoC tint — activity tint of offset+1 vs $\sigma=16250374$

Genotype fields tinted by isolated-rule activity

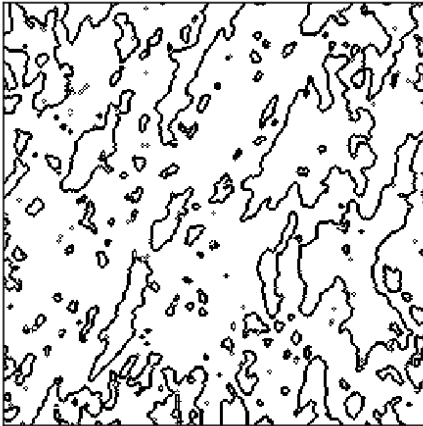


eoc_interface.png

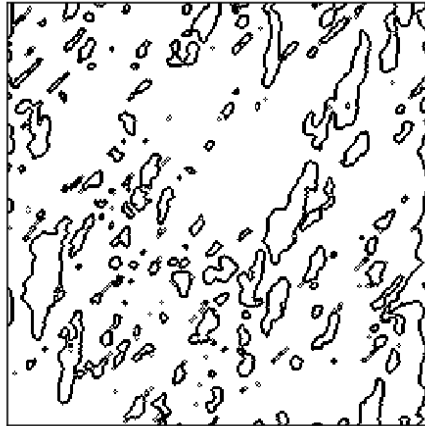
EoC interface — fractal activity-domain boundaries ($D \approx 1.6$ –1.8)

Activity-domain boundaries at threshold 0.35

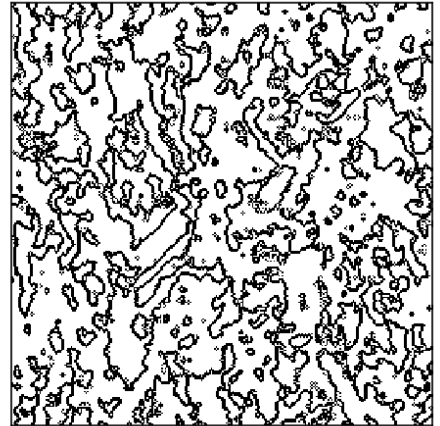
offset +1: $D=1.64$



$\sigma=16250374$: $D=1.58$



river: $D=1.80$



eoc_phase.png

EoC phase — offset+1 finite-size interrupter- q scan (smooth crossover, metastable band)

offset+1 feedforward scan across system size L (row width in cells, 30–240; 32 seeds): a smooth crossover with no critical point in the band, yet run-to-run fluctuations grow with L at $q = 1$, where system-spanning domains form (bottom: one realization across q , at $L = 240$)

